

The Impact of Patient Clinical Information and Color Constancy Preprocessing on Automated Skin Cancer Detection Using a Fusion Deep Learning Model

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Abstract

Skin cancer is among the most frequently diagnosed cancers worldwide, and the shortage of dermatologists and dermatoscopes in many regions has motivated the development of computer-aided diagnosis (CAD) systems based on ordinary clinical photographs rather than dermoscopic images. Building on the PAD-UFES-20 dataset of smartphone-acquired skin lesion images and associated patient clinical data, this work investigates whether fusing patient clinical information with image features, combined with a gray-world color constancy preprocessing step, improves automated skin cancer classification. A dual-branch deep learning model was designed, combining an EfficientNetB3 image encoder enhanced with a squeeze-and-excitation (SE) attention block with a multilayer clinical-data encoder, fused through a joint classification head. To obtain a trustworthy performance estimate, the dataset was split at the patient level using stratified group sampling so that no patient's images could appear in more than one partition, and the model was trained and evaluated using patient-level stratified 5-fold cross-validation with probability-averaging ensembling on a held-out test set. The resulting ensemble achieved 73.9% overall accuracy, 72.5% weighted precision, 73.9% weighted recall, and 72.7% weighted F1-score on 329 held-out images spanning six diagnostic categories. Performance was strong for Actinic Keratosis, Basal Cell Carcinoma, and Nevus, moderate for Seborrheic Keratosis, and considerably weaker for Squamous Cell Carcinoma and Melanoma, the two rarest classes in the dataset. These results confirm that combining clinical metadata with color-corrected images in a fused architecture, evaluated under a leakage-free patient-level protocol, provides a realistic and reasonably accurate basis for smartphone-based skin cancer screening, while highlighting class imbalance as the main remaining obstacle to further improvement.

Keywords: Skin cancer detection; Convolutional neural networks; Clinical data fusion; Color constancy; Class imbalance; PAD-UFES-20.

1. Introduction

Skin cancer is a chronic disease characterized by the abnormal growth of skin cells, most commonly triggered by ultraviolet light exposure. The World Health Organization estimates that roughly one in three cancer diagnoses worldwide is a skin cancer, and in Brazil, skin cancer alone accounts for around a third of all recorded cancer cases. The disease is broadly divided into melanoma, the rarest but most lethal form due to its high metastatic potential, and the keratinocyte cancers – basal cell carcinoma (BCC) and squamous cell carcinoma (SCC) – which are far more common but considerably less likely to metastasize.

Dermatologists typically diagnose suspicious lesions through visual inspection, often assisted by a dermatoscope, a device that reveals sub-surface colors and structures invisible to the naked eye. However, the diagnostic benefit of dermatoscopy depends strongly on the examiner's experience, and many regions – particularly in developing countries – face an acute shortage of both dermatologists and dermatoscopes. This has motivated growing interest in computer-aided diagnosis (CAD) systems that work directly from ordinary clinical photographs, ideally captured with a smartphone, so that screening is not gated behind specialized equipment.

Early CAD approaches relied on handcrafted features derived from the ABCD(E) rule or the 7-point checklist, feeding classifiers such as Support Vector Machines. These methods were designed primarily for pigmented lesions and therefore struggled with non-pigmented cancers such as BCC and SCC, and their handcrafted features generalized poorly across imaging conditions. The advent of deep learning, and convolutional neural networks (CNNs) in particular, has since become the dominant paradigm for skin lesion classification, with several studies reporting classification performance approaching or exceeding that of practicing dermatologists.

Most existing deep learning studies for skin cancer detection rely on dermoscopic images, largely because of the availability of large public dermoscopy datasets such as those released by the International Skin Imaging Collaboration (ISIC). Comparatively little work has addressed clinical (non-dermoscopic) photographs, or has combined image data with the patient clinical information – lesion location, patient age, itching, bleeding, and similar factors – that dermatologists routinely use alongside visual inspection. The PAD-UFES-20 dataset, collected through a smartphone application by the Dermatological Assistance Program (PAD) at the Federal University of Espírito Santo (UFES) in Brazil, was created specifically to close this gap by pairing clinical images with structured patient metadata.

This work builds on that dataset and on the PAD-UFES-20 baseline study by Pacheco and Krohling, which first demonstrated that combining clinical images with patient metadata through a feature-aggregation mechanism improves classification accuracy. We extend this line of work in three ways: (1) we apply a gray-world color constancy algorithm to correct for the highly variable lighting and camera conditions present in smartphone-acquired images before feature extraction; (2) we fuse image and clinical features through an attention-based architecture built on EfficientNetB3; and (3) we evaluate the resulting model under a patient-level stratified group cross-validation protocol, which prevents images of the same patient or lesion from being split across training, validation, and test sets – a leakage source that is easy to overlook given that roughly one in three lesions in this dataset has more than one associated photograph. The remainder of this paper is organized as follows: Section 2 reviews related work; Section 3 describes the dataset, preprocessing, and proposed model; Section 4 details the experimental setup; Section 5 reports the results; Section 6 discusses their implications and limitations; and Section 7 concludes the paper.

2. Related Work

Automated skin lesion classification has been studied extensively using deep learning. Yu et al. proposed a very deep CNN together with techniques for learning effectively from limited training data, while Esteva et al. fine-tuned a GoogLeNet-based CNN on more than 120,000 dermoscopic and clinical images and reported diagnostic performance comparable to board-certified dermatologists. Subsequent comparative studies by Haenssle and by Brinker likewise found that CNN-based classifiers could match, and in some cases exceed, dermatologist-level performance on curated benchmarks. Ensembling multiple CNN architectures and aggregating features across models has also been explored as a way to further improve robustness.

Most of this literature, however, is built on dermoscopic image datasets, which are more standardized and widely available than clinical photographs but require a dermatoscope to acquire – a piece of equipment that remains scarce in many low-resource settings. Brinker et al. reviewed the broader deep-learning-for-skin-cancer literature and concluded that incorporating patient clinical data alongside images tends to improve classification quality, a conclusion echoed by Kharazmi et al., who combined dermoscopic images with five clinical features to detect basal cell carcinoma specifically. That study,

however, was limited to a single cancer subtype, did not analyze how much each clinical feature actually contributed to the model's predictions, and did not consider clinical (non-dermoscopic) photographs.

The PAD-UFES-20 baseline study by Pacheco and Krohling is, to our knowledge, the first to combine clinical photographs with structured patient metadata at scale, reporting an approximately 7 percentage-point improvement in balanced accuracy when metadata was added to a CNN-only baseline. That work, however, did not investigate color constancy preprocessing, which is particularly relevant for smartphone-acquired images where lighting and camera characteristics vary widely between patients and clinics. The present study builds directly on this foundation, combining clinical data fusion with gray-world color correction, and evaluates the resulting model under a stricter, leakage-free, patient-level validation protocol than has typically been reported in this line of work.

3. Materials and Methods

3.1 Dataset

This study uses the PAD-UFES-20 dataset, collected via a smartphone application developed to support the Dermatological Assistance Program (PAD) at UFES, Brazil, a non-profit clinic providing skin lesion care to low-income patients. The application allowed physicians and medical students to photograph skin lesions and record accompanying clinical information, including the patient's age, gender, Fitzpatrick skin type, smoking and drinking history, family background, and lesion-specific attributes such as anatomical region, diameter, itching, bleeding, elevation, and whether the lesion had grown or changed. Because dermatologists in the PAD program recognize well over one hundred distinct anatomical regions, these were consolidated into a smaller, clinically meaningful set of body areas (e.g., face, scalp, neck, chest, back, arm, forearm, hand, thigh and foot) for analysis.

The working dataset used in this study contains 2,298 clinical images spanning six diagnostic categories: Basal Cell Carcinoma (BCC), Actinic Keratosis (ACK), Nevus (NEV), Seborrheic Keratosis (SEK), Squamous Cell Carcinoma (SCC), and Melanoma (MEL). Of these, BCC, SCC, and MEL are skin cancers, while ACK, NEV, and SEK are benign or pre-malignant skin conditions commonly mistaken for cancer on visual inspection. Table 1 summarizes the number of samples per category, and confirms that the dataset is considerably imbalanced – a well-known characteristic of skin lesion datasets in general, also present in the widely used ISIC dermoscopy archive.

Table 1. Number of samples per diagnostic category in the dataset used in this study.

Diagnostic category	Type	N° of images
Basal Cell Carcinoma (BCC)	Skin cancer	845
Actinic Keratosis (ACK)	Skin condition	730
Nevus (NEV)	Skin condition	244
Seborrheic Keratosis (SEK)	Skin condition	235
Squamous Cell Carcinoma (SCC)	Skin cancer	192
Melanoma (MEL)	Skin cancer	52
Total	—	2,298

An important structural property of this dataset, confirmed during data preparation, is that a substantial fraction of lesions and patients are represented by more than one photograph: 2,298 images correspond

to only 1,641 unique lesions and 1,373 unique patients. A naïve random train/test split therefore risks placing different photographs of the same lesion or patient in both the training and test sets, which would leak information across the split and artificially inflate the reported accuracy. This issue is addressed directly in the training strategy described in Section 3.7.

3.2 Exploratory Data Analysis

An exploratory analysis of the dataset was carried out to characterize the distribution of diagnostic categories and clinical attributes before model development. Figure 1 shows that BCC and ACK are the most frequent categories, while SCC and, especially, MEL are markedly underrepresented – the imbalance that motivates the class-weighting strategy described in Section 3.6.

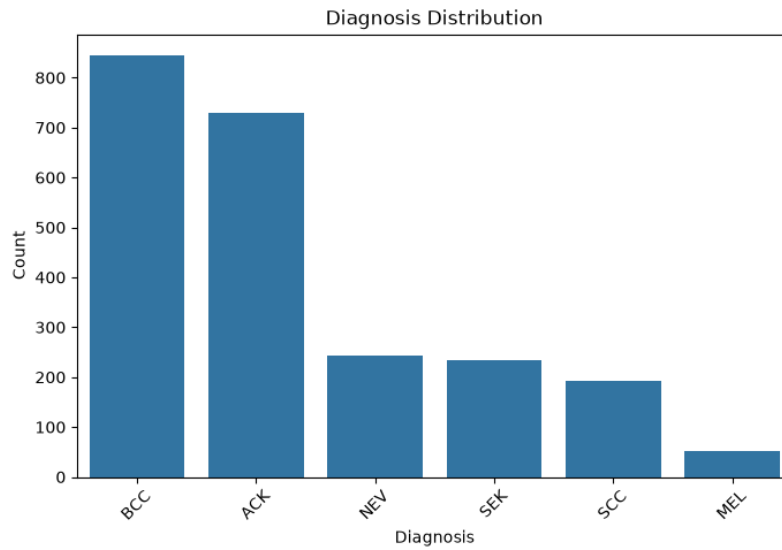


Fig. 1. Distribution of diagnostic categories in the dataset.

The age distribution of patients, shown in Figure 2, is approximately bell-shaped and centered around late middle age, consistent with the well-established association between cumulative sun exposure and skin lesion incidence.

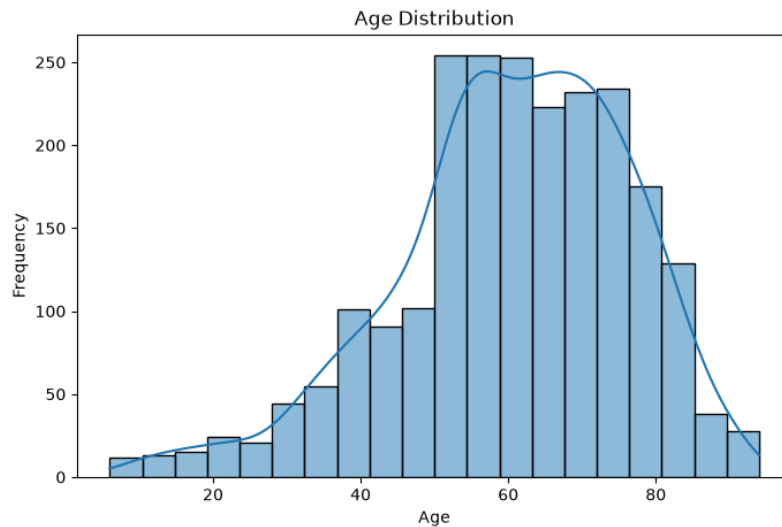


Fig. 2. Age distribution of patients in the dataset.

Figure 3 shows the gender distribution of the patients in the dataset, and Figure 4 shows the distribution of Fitzpatrick skin types, which characterize an individual's skin sensitivity to ultraviolet light and are a well-known risk factor for several of the diagnostic categories considered here.

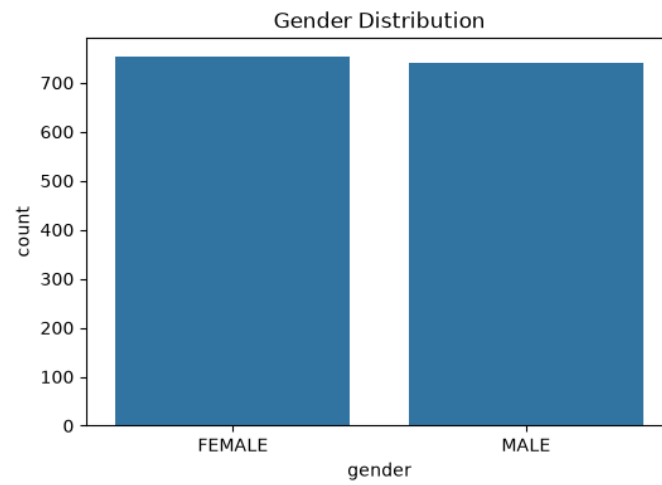


Fig. 3. Gender distribution of patients in the dataset.

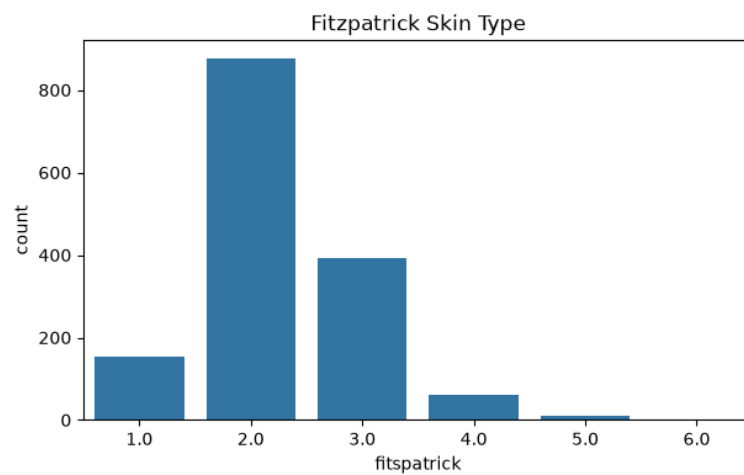


Fig. 4. Distribution of Fitzpatrick skin type among patients.

Figure 5 breaks down the diagnostic categories by patient gender, and Figure 6 shows the distribution of lesions across the consolidated body regions described in Section 3.1, both of which were used to sanity-check that no single demographic group or body region dominates any diagnostic category in a way that could bias the model.

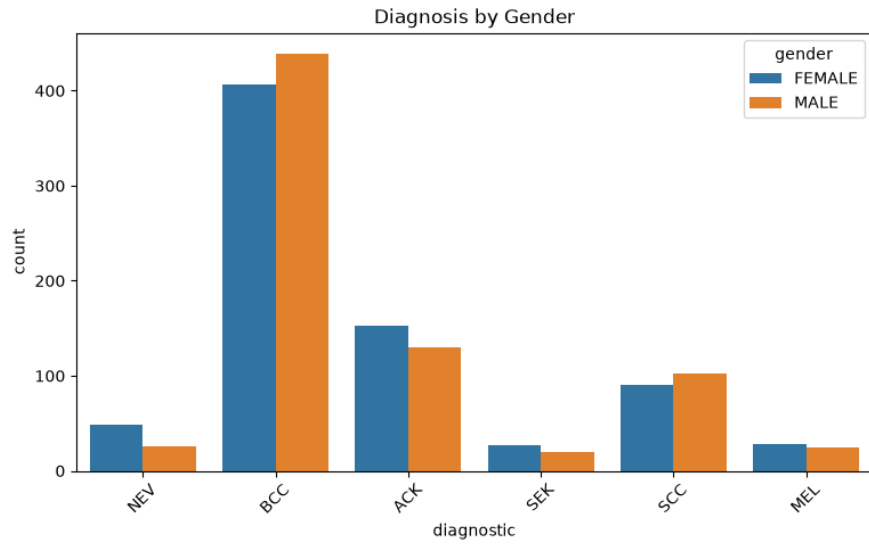


Fig. 5. *Diagnosis distribution stratified by patient gender.*

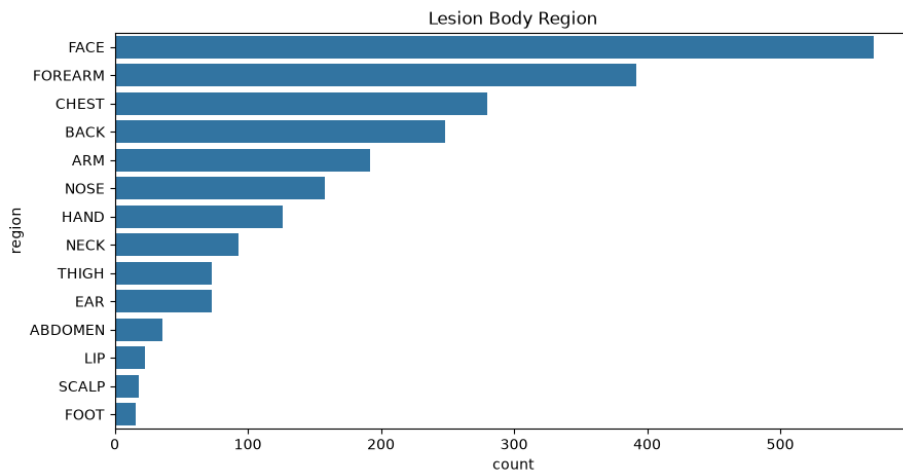


Fig. 6. *Distribution of lesion body regions in the dataset.*

Finally, Figure 7 presents the age distribution within each diagnostic category as box plots. All six categories share a similar median age around 55–60 years, but MEL shows the widest age spread, reflecting the smaller and more heterogeneous sample available for this class.

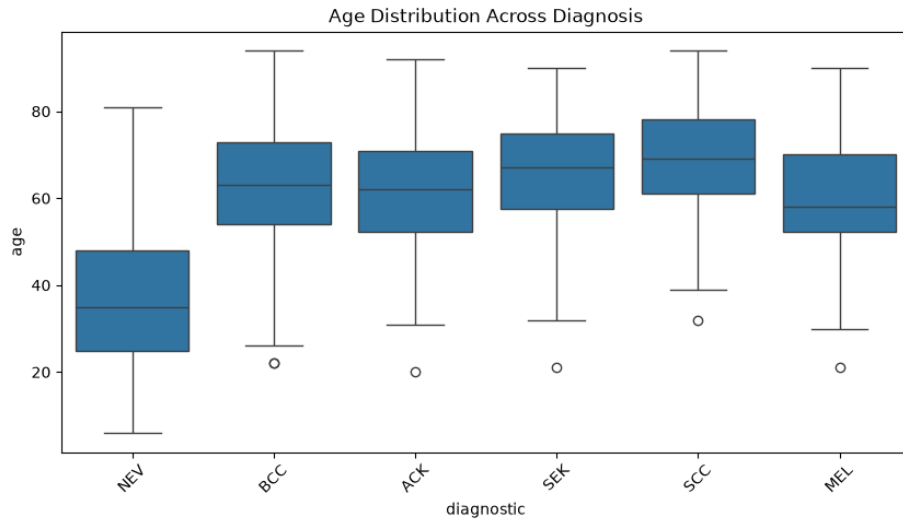


Fig. 7. Age distribution across diagnostic categories.

3.3 Color Constancy Preprocessing

Because the images in this dataset were captured with different smartphone cameras under uncontrolled lighting conditions, color inconsistencies between images can obscure the color and texture cues that are diagnostically relevant for skin lesion classification. To mitigate this, a gray-world color constancy algorithm was applied to every image prior to resizing. The gray-world assumption states that, on average, the reflectance of a natural scene is achromatic (gray); accordingly, the algorithm rescales each color channel so that its mean matches the overall gray-level mean of the image, correcting for illuminant color casts introduced by different cameras, flashes, and ambient lighting. Figure 8 illustrates the effect of this correction on a representative image from the dataset.

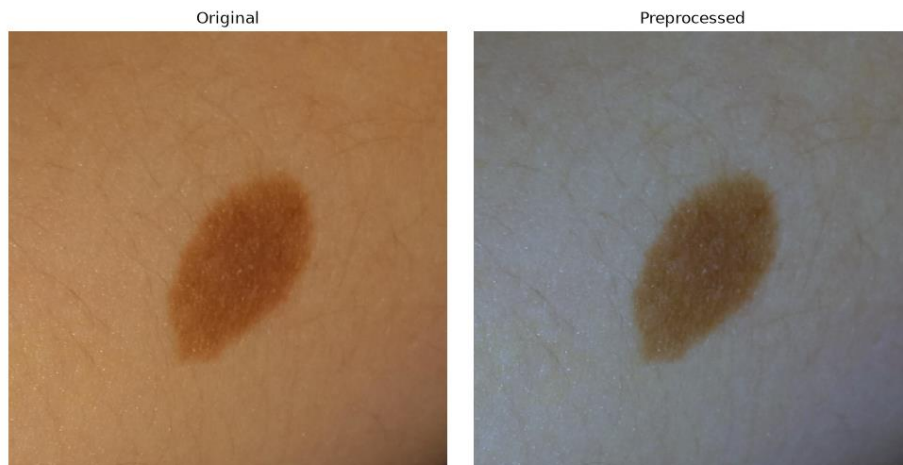


Fig. 8. Effect of gray-world color constancy preprocessing: original image (left) versus color-corrected image (right).

Beyond improving visual color stability, this preprocessing step is intended to let the downstream CNN focus on lesion-specific color and texture patterns rather than on incidental illumination artifacts, which is particularly important when combining images acquired across many different clinical settings, as is the case with smartphone-collected data. Figure 9 shows representative preprocessed images for each of the six diagnostic categories after color correction and resizing to the 300×300 input resolution used by the model.

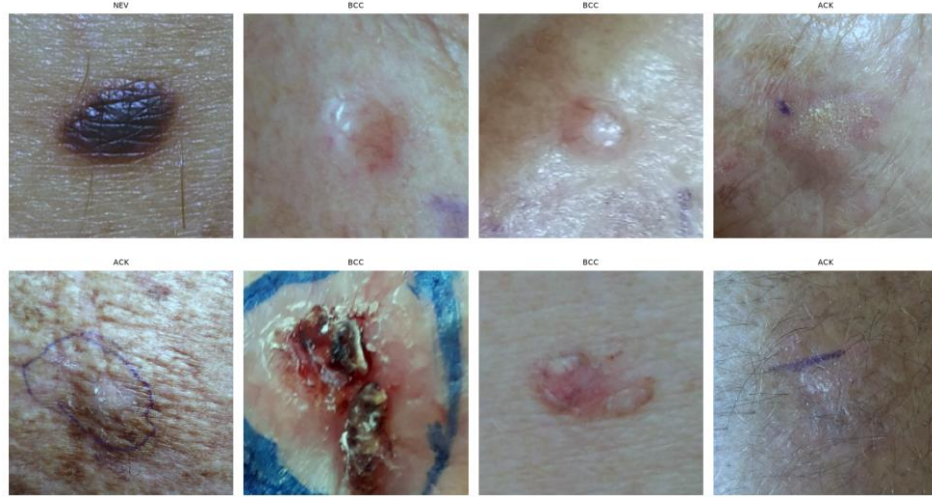


Fig. 9. Representative preprocessed samples for each diagnostic category.

3.4 Clinical Data Preprocessing

Fifteen clinical attributes were retained for modeling: patient age, gender, Fitzpatrick skin type, lesion body region, smoking and drinking history, family background (father's and mother's ethnic origin), personal and family history of skin cancer, access to piped water and a sewage system, pesticide exposure, and the two lesion diameter measurements. Binary and low-cardinality attributes (e.g., gender, smoking status) were label-encoded, while the higher-cardinality nominal attributes – body region and family background – were one-hot encoded rather than label-encoded, since label encoding would otherwise impose an artificial numerical ordering on categories that have no natural order. After encoding, the clinical feature vector has 50 dimensions. All numerical clinical features were standardized (zero mean, unit variance) using a scaler fit exclusively on each fold's training partition, to avoid leaking test-set statistics into the model, as detailed in Section 3.7.

Figure 10 shows the pairwise correlation structure of the encoded clinical features. As expected, the two lesion-diameter measurements are strongly correlated with one another, while most remaining clinical attributes are weakly correlated, indicating that they largely contribute complementary, non-redundant information to the model.

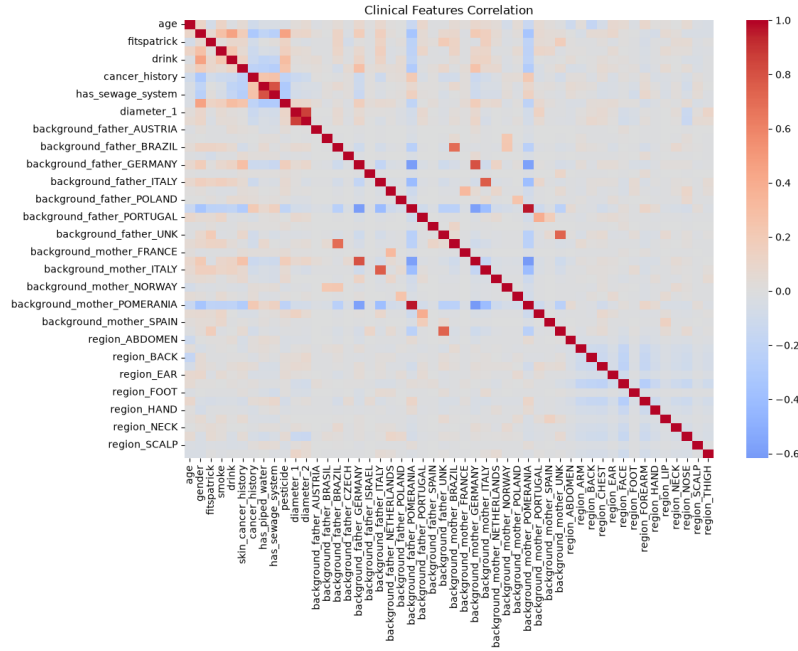


Fig. 10. Correlation heatmap of the encoded clinical features.

3.5 Proposed Fusion Architecture

A convolutional neural network (CNN) is a class of neural network specialized for extracting spatial visual features through learnable convolutional kernels, pooling layers that progressively reduce spatial resolution while controlling overfitting, and fully connected layers that perform the final classification. Building on the success of prior skin cancer studies that employed architectures such as GoogLeNet, ResNet, and VGGNet, this work uses EfficientNetB3, pretrained on ImageNet, as the image feature extractor, leveraging transfer learning to compensate for the comparatively modest size of the PAD-UFES-20 dataset relative to the millions of images typically required to train a CNN of this depth from scratch.

The proposed architecture processes each sample through two parallel branches before fusing them for classification. The image branch passes the 300×300 color-corrected image through EfficientNetB3, applies a squeeze-and-excitation (SE) attention block to the resulting feature maps – which lets the network learn to emphasize the most diagnostically informative channels rather than treating all extracted features equally – and reduces the result to a 512-dimensional feature vector through global average pooling, batch normalization, dropout, and a dense layer. The clinical branch passes the 50-dimensional clinical feature vector through a three-layer multilayer perceptron ($128 \rightarrow 64 \rightarrow 32$ units), each layer followed by batch normalization and dropout, producing a compact 32-dimensional clinical embedding. A second SE attention gate is applied to the image embedding before the two branches are concatenated, and the resulting joint representation is passed through a further stack of fully connected layers ($512 \rightarrow 256 \rightarrow 128$ units, each with L2 regularization, batch normalization, and dropout) before a final softmax layer produces the probability distribution over the six diagnostic categories.

This fusion strategy allows the classifier to weigh visual and clinical evidence jointly, mirroring how a dermatologist combines what they see in a lesion with what they know about the patient, rather than relying on either source of information in isolation.

3.6 Class Imbalance Handling and Training Strategy

Given the pronounced class imbalance documented in Table 1, class weights were computed using the standard “balanced” heuristic (inversely proportional to class frequency) and then damped with a square-root transform to avoid over-correcting for the rarest classes at the expense of the majority classes, a pitfall observed during preliminary experiments where undamped weights for MEL improved MEL recall but severely degraded ACK recall. Table 2 lists the resulting per-class weights used during training.

Table 2. Damped class weights used during model training.

Category	ACK	BCC	MEL	NEV	SCC	SEK
Weight	0.72	0.67	2.70	1.25	1.41	1.28

Training followed a two-phase transfer-learning strategy. In the first phase, the EfficientNetB3 backbone was kept frozen and only the clinical branch, attention blocks, and fusion head were trained for up to 30 epochs using the Adam optimizer with a learning rate of 1×10^{-4} . In the second phase, the top 25% of the backbone’s layers were unfrozen for fine-tuning (with batch normalization layers kept frozen to preserve stable running statistics) for up to 35 further epochs at a reduced learning rate of 1×10^{-5} . Both phases used label-smoothed categorical cross-entropy loss, weighted by the class weights in Table 2, together with early stopping (monitoring validation loss), learning-rate reduction on plateau, and checkpointing of the best-performing epoch by validation accuracy. Training images were augmented on the fly with random horizontal and vertical flips, random 90-degree rotations, and random brightness jitter, reflecting the fact that skin lesion photographs have no canonical orientation. Mixed-precision (float16/float32) training was used throughout to reduce memory usage and accelerate training on GPU hardware.

3.7 Patient-Level Stratified Group Cross-Validation and Ensemble Evaluation

To obtain a trustworthy performance estimate given the patient/lesion duplication described in Section 3.1, all data splits in this study were performed using stratified group k-fold sampling, grouping by patient identifier rather than sampling images independently. A held-out test set of 329 images was first set aside using this procedure, guaranteeing that no patient appearing in the test set contributes any image to model training. The remaining 1,969 images were then partitioned into five stratified, patient-grouped folds. For each fold, a fresh instance of the fusion model described in Section 3.5 was trained from scratch using the two-phase strategy of Section 3.6, with the other four folds serving as that model’s training data and the held-out fold serving as its validation set; the standardization scaler for clinical features was likewise refit on each fold’s own training partition only.

At inference time, each of the five fold-specific models produced a softmax probability distribution over the six diagnostic categories for every image in the held-out test set, and these five probability vectors were averaged to obtain the final ensemble prediction. This design serves two purposes: it yields a performance estimate free of patient-level leakage at every stage of the pipeline, and it exploits the variance-reduction benefit of ensembling five independently trained models, which is expected to outperform any single fold model evaluated in isolation.

4. Experimental Setup

All models were implemented in TensorFlow/Keras. Images were resized to 300×300 pixels to match the native input resolution of EfficientNetB3, and training used a batch size of 32. The Adam optimizer was used throughout, with learning rates of 1×10^{-4} during the frozen feature-extraction phase and 1×10^{-5} during fine-tuning, as described in Section 3.6. Early stopping used a patience of six epochs on validation loss, and the learning rate was reduced by a factor of 0.3 after three epochs without improvement, down to a floor of 1×10^{-7} . Five models were trained in total, one per cross-validation fold, and evaluated jointly as an ensemble on the single held-out test set of 329 images described in Section 3.7.

5. Results

Table 3 reports the test accuracy achieved by each of the five fold-specific models individually, as well as by the probability-averaged ensemble, on the held-out test set.

Table 3. Per-fold and ensemble accuracy on the held-out test set (329 images).

Model	Test accuracy
Fold 0	0.7477
Fold 1	0.6717
Fold 2	0.7234
Fold 3	0.7112
Fold 4	0.6960
Ensemble (5-fold average)	0.7386

The ensemble (73.9%) outperforms every individual fold model, including the best-performing single fold (Fold 0, 74.8% – within one point of the ensemble but not consistently reproduced across folds), confirming that averaging the five fold-specific models reduces variance and yields a more reliable predictor than any single model trained on one partition of the data. Table 4 summarizes the overall weighted performance metrics of the ensemble.

Table 4. Overall ensemble performance on the held-out test set.

Metric	Value
Accuracy	0.7386
Precision (weighted)	0.7247
Recall (weighted)	0.7386
F1-score (weighted)	0.7274
F1-score (macro)	0.6243

"Table 5 breaks these results down by diagnostic category. The model performs strongly on ACK, BCC, and NEV, the three best-represented categories, and reasonably well on SEK. Performance on SCC and MEL, the two rarest categories, is considerably weaker, with SCC in particular showing low precision and recall despite the elevated class weight assigned to it during training.

Table 5. Per-class precision, recall, and F1-score of the ensemble model on the held-out test set.

Category	Precision	Recall	F1-score	Support
ACK	0.82	0.80	0.81	104
BCC	0.72	0.86	0.78	121
MEL	0.50	0.43	0.46	7
NEV	0.84	0.77	0.81	35
SCC	0.31	0.18	0.23	28
SEK	0.70	0.62	0.66	34

Figure 11 presents the confusion matrix of the ensemble model on the held-out test set, showing that most SCC misclassifications are attributed to ACK and BCC, the two categories with which it shares the greatest visual similarity, consistent with SCC being one of the three non-pigmented diagnostic categories in this dataset alongside ACK and BCC.

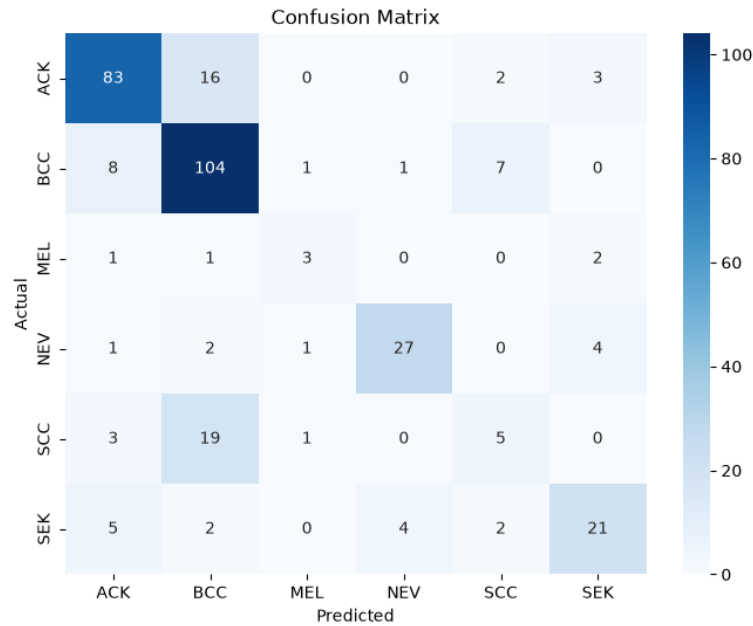


Fig. 11. Confusion matrix of the ensemble model on the held-out test set.

Figure 12 shows the per-class receiver operating characteristic (ROC) curves and corresponding area-under-the-curve (AUC) values, computed from the ensemble's averaged class probabilities. Consistent with Table 5, the categories with the highest AUC are ACK, BCC, and NEV, while SCC shows the lowest ranking quality among the six categories.

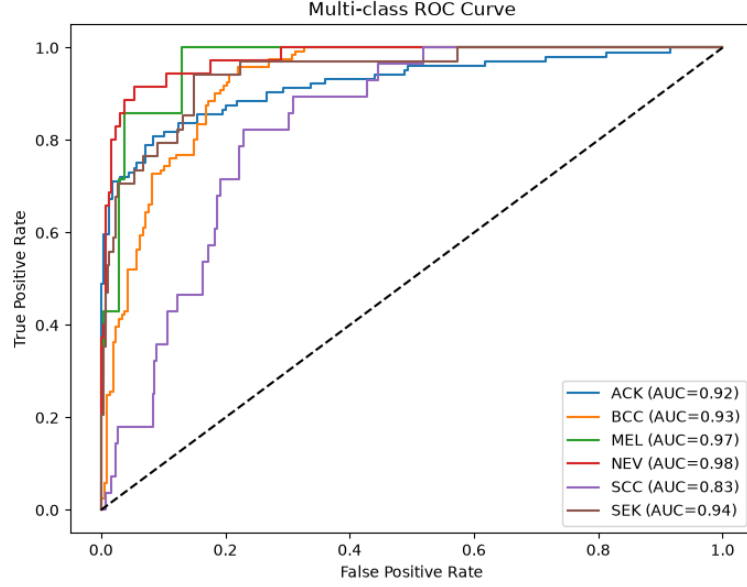


Fig. 12. Per-class ROC curves of the ensemble model on the held-out test set.

6. Discussion

The results confirm that fusing patient clinical information with color-corrected image features, evaluated under a leakage-free patient-level protocol, yields a model capable of distinguishing the six PAD-UFES-20 diagnostic categories with an overall accuracy of 73.9% and a weighted F1-score of 72.7%. The clear gap between weighted and macro-averaged F1-score (0.727 versus 0.624) quantifies the extent to which performance is driven by the majority classes; the model is considerably more reliable for ACK, BCC, and NEV than for the rarer SCC and MEL categories.

The persistently weak performance on SCC, which was observed consistently across several loss functions and class-weighting configurations explored during model development, appears to be driven primarily by two factors: the limited number of SCC training examples relative to BCC and ACK, and genuine visual overlap between SCC and these two non-pigmented categories, which the confusion matrix in Figure 11 corroborates directly. This is consistent with previously reported difficulties in distinguishing keratinocyte cancers from benign keratoses in the dermatology literature, and suggests that further gains for this category are more likely to come from targeted data collection or class-specific augmentation than from further tuning of the loss function alone. The very small number of MEL test samples (7 images) similarly limits how much can be concluded about melanoma-specific performance from this study alone, despite melanoma being clinically the most dangerous category.

Compared to the original PAD-UFES-20 baseline study, which reported an approximately 7 percentage-point improvement in balanced accuracy from adding clinical metadata to an image-only CNN, a direct numerical comparison with the results reported here is not appropriate, since the present study uses a different, more conservative patient-level evaluation protocol specifically designed to eliminate the leakage risk created by patients and lesions with multiple photographs. We view this stricter protocol, together with the 5-fold ensembling strategy, as a methodological contribution in its own right:

it demonstrates that a clinically meaningful accuracy level can be reached even under an evaluation regime that is deliberately harder to inflate than a simple random split.

This study has several limitations. First, only a single backbone architecture (EfficientNetB3) was evaluated; comparing against alternative backbones, or an ensemble across architectures rather than across folds of the same architecture, could further clarify how much of the remaining error is attributable to the specific CNN used. Second, the severe imbalance in the underlying dataset, particularly for MEL, limits the statistical confidence of per-class conclusions for the rarest categories regardless of modeling choices. Third, while the gray-world color constancy step is intended to normalize illumination differences across the heterogeneous smartphone images in this dataset, this study did not isolate its individual contribution through a controlled ablation against an otherwise identical model trained without color correction; quantifying that specific contribution is a natural next step.

7. Conclusion

This work presented a fusion deep learning model that combines gray-world color-corrected clinical images with patient clinical data for automated skin cancer classification on the PAD-UFES-20 dataset, using an EfficientNetB3 backbone with squeeze-and-excitation attention and a dedicated clinical-data encoding branch. To obtain a realistic performance estimate, all data splits were performed at the patient level using stratified group sampling, and the final model was evaluated as a five-fold cross-validation ensemble on a held-out test set never seen during training. The resulting ensemble achieved 73.9% accuracy and a weighted F1-score of 72.7%, with strong performance on the more common diagnostic categories (ACK, BCC, NEV) and a clear, consistently reproduced weakness on the rarest category, SCC. These findings support the value of combining clinical metadata with color-normalized images for smartphone-based skin cancer screening, while underscoring that addressing class imbalance – through targeted data collection, augmentation, or loss design – remains the most promising direction for further improving performance on the rarer cancer subtypes.

References

This manuscript refers throughout to prior work by Pacheco and Krohling (the PAD-UFES-20 baseline study), Esteva et al., Yu et al., Haenssle et al., Brinker et al., and Kharazmi et al. Full bibliographic entries for these and any additional sources should be inserted here in the citation style required by the target venue before submission.